

AI Customer Support Agent – Validation & Test Set

Functional validation of the working MVP

Purpose

This document records the functional validation performed on the AI Customer Support Agent MVP. The tests cover knowledge-grounded policy questions, order operations, support-ticket workflows, human escalation, and an unsupported/uncertain knowledge case.

Validation Environment

Frontend: Streamlit | Chat model: llama3.1:8b | Embedding model: nomic-embed-text:latest | Retrieval: Chroma/RAG | Operational data: SQLite | Execution: local development environment

Test Summary

Metric	Result
Total test cases	9
Passed	9
Failed	0
Pass rate	100%

Detailed Test Cases

ID	Scenario	Input	Expected Result	Actual Result	Status
TC-01	Return policy	What is the return policy?	System should provide the applicable return window and eligibility information from the knowledge base.	Return requests are accepted within 30 calendar days from delivery, subject to eligibility requirements.	PASS
TC-02	Return timeframe reasoning	Can I return an item within 10 days?	System should correctly understand that 10 days falls within a 30-day return window and mention eligibility conditions.	System correctly identified 10 days as within the 30-day window and explained eligibility requirements.	PASS
TC-03	Non-returnable items	What items cannot be returned?	System should list non-returnable categories from the knowledge base.	Personalized products, gift cards, downloadable digital products, and final-sale products were identified as non-returnable.	PASS
TC-04	Order lookup – shipped	Where is my order NC1002?	System should retrieve the order details and current status.	NC1002: Smart Watch, \$3999.00, status Shipped.	PASS
TC-05	Order lookup – delivered	Where is my order NC1001?	System should retrieve the order details, delivery status, and delivery date when available.	NC1001: Wireless Headphones, \$2499.00, status Delivered, delivered 2026-08-10.	PASS
TC-06	Damaged-order ticket creation	My order NC1001 arrived damaged. Please create a support ticket.	System should create a support ticket for the damaged-order issue and return the ticket information.	A support ticket was created successfully with an automatically generated ticket ID and Open status.	PASS

ID	Scenario	Input	Expected Result	Actual Result	Status
TC-07	Ticket status lookup	What is the status of TKT0018?	System should retrieve the status and details of the specified ticket.	TKT0018 was returned with status Open, issue 'NC1001 arrived damaged; create ticket', and order ID NC1001.	PASS
TC-08	Human escalation	I want to speak to a human agent.	System should create an escalation ticket and provide its status.	Human support escalation was created successfully; example result TKT0016 with status Open.	PASS
TC-09	Unsupported knowledge question	Does the company deliver to Antarctica?	System should not invent a delivery policy when the knowledge base does not provide sufficient information.	System stated that it did not have enough information in the knowledge base to answer specifically about Antarctica delivery.	PASS

Key Validation Observations

- The final 8B chat model correctly handled the return-policy timeframe reasoning that previously produced an incorrect result with the smaller 3B model.
- Order lookup returned the expected operational details for both shipped and delivered orders.
- Support-ticket creation and ticket-status lookup worked with automatically generated ticket IDs.
- Human escalation successfully created an Open support ticket.
- The unsupported Antarctica delivery question was handled without inventing a company-specific policy.

Conclusion

All nine representative validation scenarios passed. The working MVP demonstrated successful knowledge-grounded answering, controlled operational actions, ticket workflows, human escalation, and uncertainty handling for unsupported knowledge-base questions.